OPERATIONS MANAGEMENT

OPERATIONS PLANNING & SCHEDULING

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Operations Planning & Scheduling

- Managing supply chains effectively requires more than just good demand forecasts. Demand is the first half of the equation, and the other half is supply. The firm must develop plans to supply the resources needed to meet the forecasted demand. These resources include the work-force, materials, inventories, finance, and equipment capacity.

- **Operations planning and scheduling** is the process of making sure that demand and supply plans are in balance, from the aggregate level down to the short-term scheduling level.

- As we see with an airline example, the process begins at the aggregate level and gets progressively more specific until all crew members know their tour of duty.

- Operations planning and scheduling lies at the core of supply chain integration, around which plans are made up and down the supply chain, from supplier deliveries to customer due dates and services.

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- Operations planning and scheduling is meaningful for each organization along the supply chain. First, it requires managerial inputs from all of the firm's functions. Marketing provides inputs on demand and accounting provides important cost data and a firm's financial condition. Second, each function is affected by the plan. A plan that calls for expanding the workforce has a direct impact on the hiring and training requirements for the human resources function. As the plan is implemented, it creates revenue and cost streams that finance must deal with as it manages the firm's cash flows. Third, each department and group in a firm has its own workforce. Managers of these departments must make choices on hiring, overtime, and vacations.

- Scheduling is important for both service and manufacturing processes. Whether the business is an airline, hotel, computer manufacturer, or university, schedules are a part of everyday life. Schedules involve an enormous amount of detail and affect every process in a firm. For example, service, product, and employee schedules determine specific cash flow requirements, trigger the firm's billing process, and initiate requirements for the employee training process. Firms use the scheduling process to lower their costs and improve their responsiveness, affecting operations up and down the supply chain worldwide.

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Types of Plans with Operations Planning & Scheduling

Key Term	Definition
Sales and operations plan (S&OP)	A plan of future aggregate resource levels so that supply is in balance with demand. It states a company's or department's production rates, workforce levels, and inventory holdings that are consistent with demand forecasts and capacity constraints. The S&OP is time-phased, meaning that it is projected for several time periods (such as months or quarters) into the future.
Aggregate plan	Another term for the sales and operations plan.
Production plan	A sales and operations plan for a *manufacturing firm* that centers on production rates and inventory holdings.
Staffing plan	A sales and operations plan for a *service firm*, which centers on staffing and on other human resource-related factors.
Resource plan	An intermediate step in the planning process that lies between S&OP and scheduling. It determines requirements for materials and other resources on a more detailed level than the S&OP. It is covered in the next chapter.
Schedule	A detailed plan that allocates resources over shorter time horizons to accomplish specific tasks.

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Stages in Operations Planning & Scheduling

I Aggregation

- The sales and operations plan is useful because it focuses on a general course of action, consistent with the company's strategic goals and objectives, without getting bogged down in details.

- We must first aggregate, and then use the targets and resources from the plan to create effective, coordinated schedules. In general, companies perform aggregation along three dimensions:

1. **Product Families:** A group of customers, services, or products that have similar demand requirements and common process, workforce, and materials requirements is called a product family. For instance, a manufacturer of bicycles that produces 12 different models of bikes might divide them into two groups, mountain bikes and road bikes.

2. **Workforce:** A company can aggregate its workforce in various ways as well, depending on its flexibility. For example, if workers at the bicycle manufacturer are trained to work on either mountain bikes or road bikes, for planning purposes management can consider its workforce to be a single aggregate group, even though the skills of individual workers may differ.

3. **Time:** The planning horizon covered by a sales and operations plan typically is one year, although it can differ in various situations. To avoid the expense and disruptive effect of frequent changes in output rates and the workforce, adjustments usually are made monthly or quarterly.

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II Relationship of Operations Plans and Schedules to Other Plans

- A financial assessment of the organization's near future—that is, for 1 or 2 years ahead — is called either a business plan (in for-profit firms) or an annual plan (in for-nonprofit service organizations).

- A **business plan** is a projected statement of income, costs, and profits. It usually is accompanied by budgets, a projected (pro forma) balance sheet, and a projected cash flow statement, showing sources and allocations of funds.

- The following figure illustrates the relationships among the business or annual plan, sales & operations plan, detailed plans and schedules derived from it.

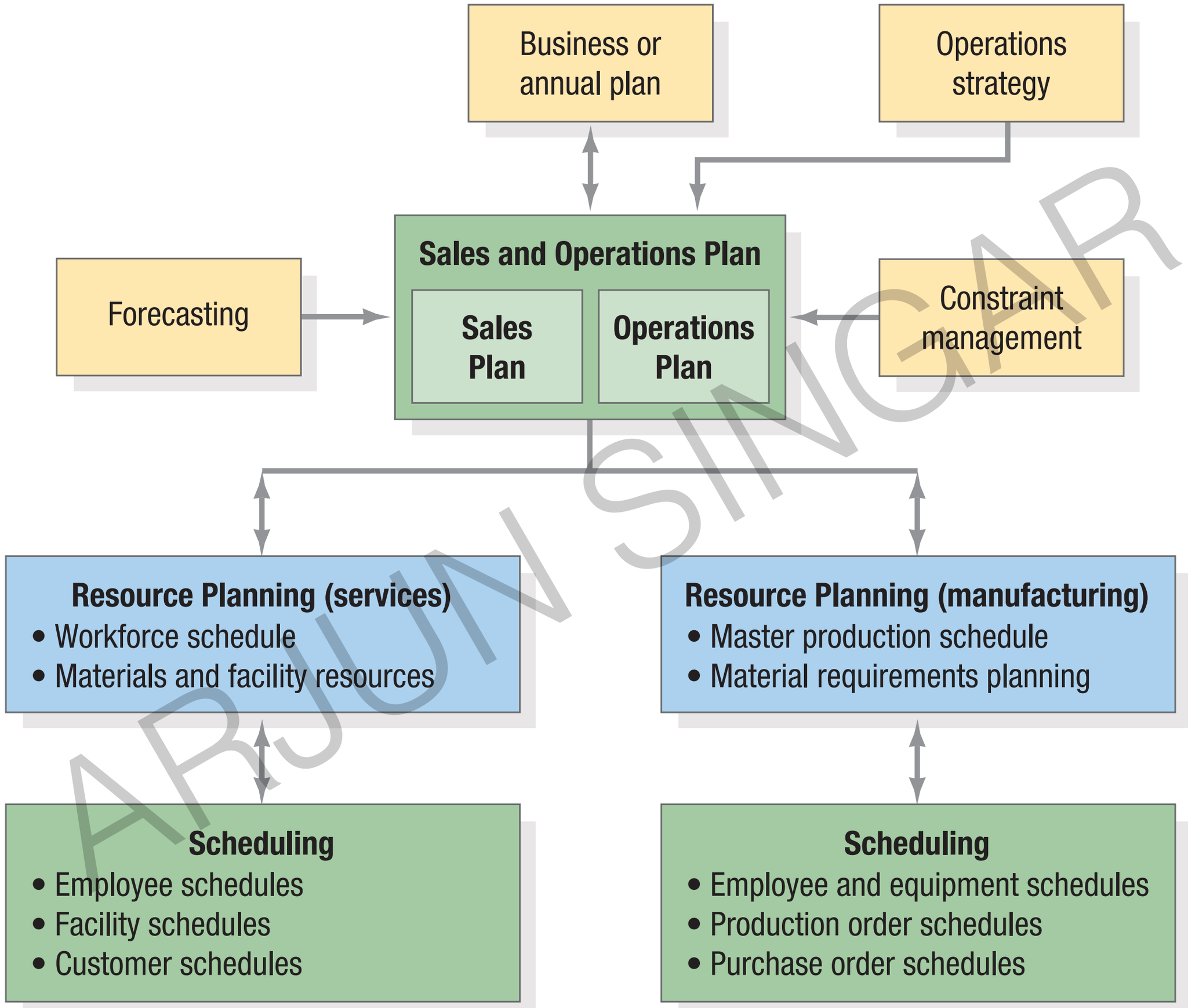
- For **service providers** (in for-profit firms) in the supply chain, top management sets the organization's direction and objectives in the business plan. This plan then provides the framework for developing the sales and operations plan, which typically focuses on staffing and other human resource-related factors at a more aggregate level. It presents the number and types of employees needed to meet the objectives of the business plan.

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- Based on the sales and operations plan for a service provider, the next planning level is ***resource planning*** to determine the firm's workforce schedules and other resource requirements, such as materials and facilities, on a more detailed level. The ***workforce schedule*** details the specific work schedule for each category of employee. For example, a sales and operations plan might allocate 10 police officers for the day shift in a particular district; the workforce schedule might assign 5 of them to work Monday through Friday and the other 5 to work Wednesday through Sunday to meet the varying daily needs for police protection in that district. The lowest planning level is ***scheduling***, which puts together day-to-day schedules for individual employees and customers.

- For **manufacturing firms** in the supply chain, top management sets the company's strategic objectives for at least the next year in the business plan. It provides the overall framework, along with inputs coming from operations strategy, forecasting, and capacity constraint management. The sales and operations plan specifies product family production rates, inventory levels, and workforce levels. The next planning level beneath the sales and operations plan is resource planning. **Resource planning** gets specific as to individual products within each product family, purchased materials, and resources on a detailed level. The ***master production schedule*** specifies the timing and size of production quantities for each product in the product families. The ***material requirements planning*** process then derives plans for components, purchased materials, and workstations. It puts together day-to-day schedules for employees, equipment, and production or purchase orders. Thus, the sales and operations plan plays a key role in translating the strategies of the business plan into an operational plan for the manufacturing 7 process.

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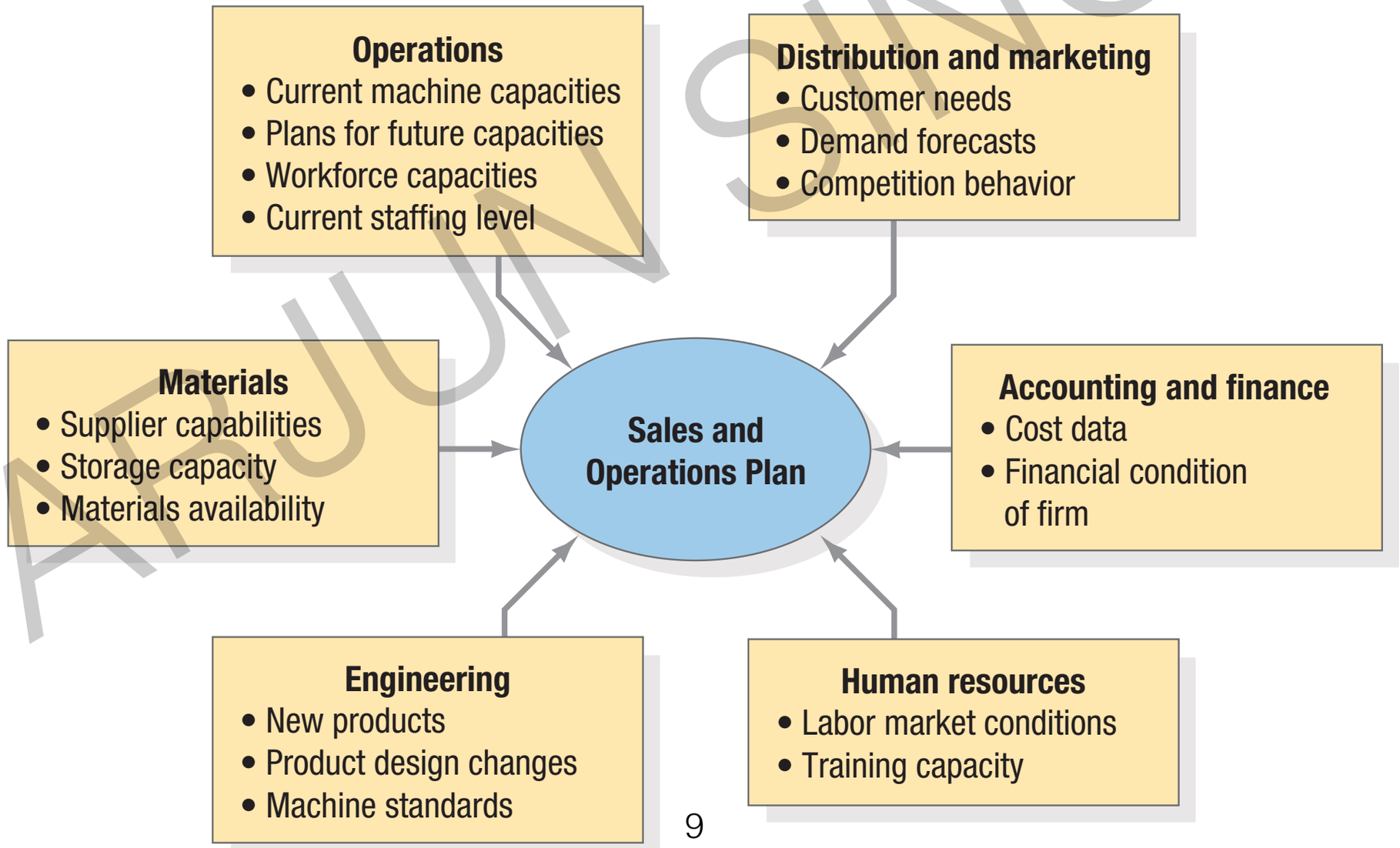
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Developing Sales & Operations Plans

- Developing sales and operations plans means making decisions. In this section, we concentrate on the information inputs, the supply options themselves, and strategies that go into the sales and operations planning (S&OP) decisions.

Information Inputs

- Just as it is needed to manage the demand side, consensus is needed among the firm's departments when decisions for the supply side are made. Information inputs are sought to create a plan that works for all. The following figure lists inputs from each functional area. Such coordination helps synchronize the flow of services, materials, and information through the supply chain to best balance supply with customer demand.



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Supply Options

- **Anticipation Inventory:** This can be used to absorb uneven rates of demand or supply. For example, a plant facing seasonal demand can stock anticipation inventory during light demand periods and use it during heavy demand periods. Manufacturers of air conditioners, such as Whirlpool, can experience 90 percent of their annual demand during just three months of a year. Extra, or anticipation inventory, also can help when supply, rather than demand, is uneven. For example, a company can stock up on a certain purchased item if the company's suppliers expect severe capacity limitations. Despite its advantages, anticipation inventory can be costly to hold, particularly if stocked in its finished state. Moreover, when services or products are customized, anticipation inventory is not usually an option.

- **Workforce Adjustment:** Management can adjust workforce levels by hiring or laying off employees. The use of this alternative can be attractive if the workforce is largely unskilled or semiskilled and the labor pool is large. These conditions are more likely found in some countries than in others. However, for a particular company, the size of the qualified labor pool may limit the number of new employees that can be hired at any one time. Also, new employees must be trained, and the capacity of the training facilities themselves might limit the number of new hires at any one time.

- **Workforce Utilization:** An alternative to a workforce adjustment is a change in workforce utilization involving overtime and undertime. **Overtime** means that employees work longer than the regular workday or workweek and receive additional pay for the extra hours. It can be used to satisfy output requirements that cannot be completed on regular time. Overtime is expensive (typically 150 percent of the regular-time pay rate) and workers often do not want to work a lot of overtime for an extended period of time. Excessive overtime also can result in declining quality and productivity. On the other hand, it helps avoid the costly fringe benefits (such as health insurance, dental care, Social Security, retirement funds, paid vacations, and holidays) that come with hiring a new full-time employee.

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- **Undertime** means that employees do not have enough work for the regular-time workday or workweek. For example, they cannot be fully utilized for eight hours per day or for five days per week. Undertime occurs when labor capacity exceeds demand requirements, and this excess capacity cannot be used productively to build up inventory or to satisfy customer orders earlier than the delivery dates already promised. Undertime can either be paid or unpaid. An example of *paid undertime* is when employees are kept on the payroll rather than being laid off. In this scenario, employees work a full day and receive their full salary but are not as busy because of the light workload. Some companies use paid undertime (though they do not call it that) during slack periods, particularly with highly skilled, hard-to-replace employees or when there are obstacles to laying off workers. The disadvantages of paid undertime include the cost of paying for work not performed and lowered productivity.

- **Part-Time Workers:** Another option apart from undertime is to hire part-time workers, who are paid only for the hours and days worked. Perhaps they only work during the peak times of the day or peak days of the week. Sometimes, part-time arrangements provide predictable work schedules, but in other cases workers are not called in if the workload is light. Such arrangements are more common in low-skill positions or when the supply of workers seeking such an arrangement is sufficient. Part-time workers typically do not receive fringe benefits. **Subcontractors** can be used to overcome short-term capacity shortages, such as during peaks of the season or business cycle.

- **Vacation Schedules:** A manufacturer can shut down during an annual lull in sales, leaving a skeleton crew to cover operations and perform maintenance. Hospital employees might be encouraged to take all or part of their allowed vacation time during slack periods. The use of this alternative depends on whether the employer can mandate the vacation schedules of its employees. In any case, employees may be strongly discouraged from taking vacations during peak periods or encouraged to take vacations during slack periods.

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Planning Strategies

Here we focus on supply options that define output rates and workforce levels. Two basic strategies are useful starting points in searching for the best plan.

- **Chase Strategy:** The chase strategy involves hiring and laying off employees to match the demand forecast over the planning horizon. Varying the workforce's regular-time capacity to equate supply to demand requires no inventory investment, overtime, or undertime. The drawbacks are the expense of continually adjusting workforce levels, the potential alienation of the workforce, and the loss of productivity and quality because of constant changes in the workforce.

- **Level Strategy:** The level strategy involves keeping the workforce constant (except possibly at the beginning of the planning horizon). It can vary its utilization to match the demand forecast via overtime, undertime (paid or unpaid), and vacation planning (i.e., paid vacations when demand is low). A constant workforce can be sized at many levels: Managers can choose to maintain a large workforce so as to minimize the planned use of overtime during peak periods (which, unfortunately, also maximizes the need for undertime during slack periods). Alternately, they can choose to maintain a smaller workforce and rely heavily on overtime during the peak periods (which places a strain on the workforce and endangers quality).

These two **"pure"** strategies used alone usually do not produce the best sales and operations plan. It might not be best to keep the workforce exactly level, or to vary it to exactly match forecasted demand on a period-by-period basis. The best strategy, therefore, usually is a **mixed strategy** that considers the full range of supply options. The mixed strategy opens things up to all options, including anticipation inventory, part-time workers, subcontractors, backorders, and stockouts.

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Constraints and Costs

- An acceptable sales and operations plan must recognize relevant constraints or costs. Constraints can be either physical limitations or related to managerial policies. Examples of physical constraints might be machine capacities that limit maximum output or inadequate inventory storage space. Policy constraints might include limitations on the number of backorders or the use of subcontractors or overtime, as well as the minimum inventory levels needed to achieve desired safety stocks. Ethical issues may also be involved, such as excessive layoffs or required overtime.

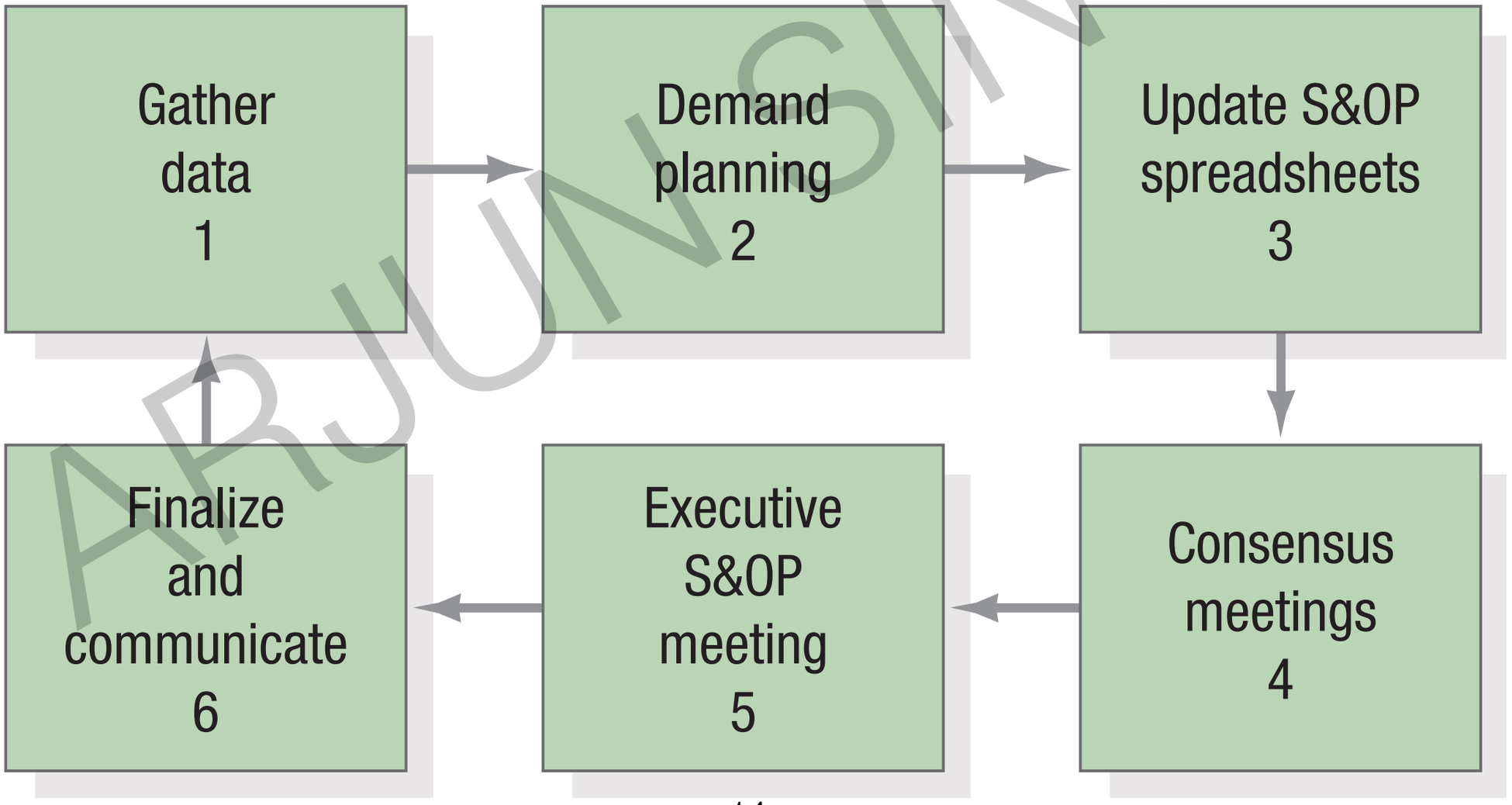
Types of Costs with S&OP

Cost	Definition
Regular time	Regular-time wages paid to employees plus contributions to benefits, such as health insurance, dental care, Social Security, retirement funds, and pay for vacations, holidays, and certain other types of absences.
Overtime	Wages paid for work beyond the normal workweek, typically 150 percent of regular-time wages (sometimes up to 200 percent for Sundays and holidays), exclusive of fringe benefits. Overtime can help avoid the extra cost of fringe benefits that come with hiring another full-time employee.
Hiring and layoff	Costs of advertising jobs, interviews, training programs for new employees, scrap caused by the inexperience of new employees, loss of productivity, and initial paperwork. Layoff costs include the costs of exit interviews, severance pay, retaining and retraining remaining workers and managers, and lost productivity.
Inventory holding	Costs that vary with the level of inventory investment: the costs of capital tied up in inventory, variable storage and warehousing costs, pilferage and obsolescence costs, insurance costs, and taxes.
Backorder and stockout	Additional costs to expedite past-due orders, the costs of lost sales, and the potential cost of losing a customer to a competitor (sometimes called loss of goodwill).

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Sales & Operations Planning as a Process

- Sales and operations planning is a decision-making process, involving both planners and management. It is dynamic and continuing, as aspects of the plan are updated periodically when new information becomes available and new opportunities emerge. It is a cross-functional process that seeks a set of plans that all of a firm's functions can support. For each product family, decisions are made based on cost trade-offs, recent history, recommendations by planners and middle management, and the executive team's judgment.



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Artic Air Company—April Sales and Operations Plan

Family: Medium window units (make-to-stock) *Unit of measure:* 100 units

	HISTORY													
SALES	J	F	M	A*	M	J	J	A	S	3rd 3 Mos**	4th 3 Mos	Mos 13–18	Fiscal Year Projection ($000)	Business Plan ($000)
New forecast	45	55	60	70	85	95	130	110	70	150	176	275	$8,700	$8,560
Actual sales	52	40	63											
Diff for month	7	–15	3											
Cum		–8	–5											
OPERATIONS														
New Plan	75	75	75	75	75	85	85	85	75	177	225			
Actual	75	78	76											
Diff for month	0	3	1											
Cum		3	4											
INVENTORY														
Plan	85	105	120	125	115	105	60	35	40	198	321			
Actual	92	130	143											

DEMAND ISSUES AND ASSUMPTIONS

1. New product design to be launched in January of next year.

SUPPLY ISSUES

1. Vacations primarily in November and December.
2. Overtime in July–August.

* April is the first month of the planning horizon for this current plan. When next month's plan is developed, its first month in the planning horizon will be May, and the most recent month of the history will be April (with January no longer shown in the history).

** This column provides the sales, operations, and inventory totals for October through December. For example, the forecast of 150 units translates into an average of 50 units per month (or 150/3 = 50).

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Example: Using the Chase and Level Strategies as Starting Points

The manager of a large distribution center must determine how many part-time stockpickers to maintain on the payroll. She wants to develop a staffing plan that minimizes total costs, and wants to begin with the chase strategy and level strategy. For the level strategy, she wants to first try the workforce level that meets demand with the minimum use of undertime and not consider vacation scheduling.

First, the manager divides the next year into six time periods, each one 2 months long. Each part-time employee can work a maximum of 20 hours per week on regular time, but the actual number can be less. Instead of paying undertime, each worker's day is shortened during slack periods. Once on the payroll, each worker is used each day, but they may work only a few hours. Overtime can be used during peak periods.

The distribution center's forecasted demand is shown as the number of part-time employees required for each time period at the maximum regular time of 20 hours per week. For example, in period 3, an estimated 18 part-time employees working 20 hours per week on regular time will be needed.

	1	2	3	4	5	6	Total
Forecasted demand*	6	12	18	15	13	14	78

*Number of part-time employees

Currently, 10 part-time clerks are employed. They have not been subtracted from the forecasted demand shown. Constraints and cost information are as follows:

a. The size of training facilities limits the number of new hires in any period to no more than 10.

b. No backorders are permitted; demand must be met each period.

c. Overtime cannot exceed 20 percent of the regular-time capacity (that is, 4 hours) in any period. Therefore, the most that any part-time employee can work is 1.20(20) = 24 hours per week.

d. The following costs can be assigned:

Regular-time wage rate	$2,000 per time period at 20 hours per week
Overtime wages	150 percent of the regular-time rate
Hires	$1,000 per person
Layoffs	$500 per person

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a. Chase Strategy

This strategy simply involves adjusting the workforce as needed to meet demand, as shown in Figure 15.5. Rows in the spreadsheet that do not apply (such as inventory and vacations) are hidden. The workforce level row is identical to the forecasted demand row. A large number of hirings and layoffs begin with laying off four part-time employees immediately because the current staff is 10 and the staff level required in period 1 is only six. However, many employees, such as college students, prefer part-time work. The total cost is $173,500, and most of the cost increase comes from frequent hiring and layoffs, which add $17,500 to the cost of utilized regular-time costs.

	1	2	3	4	5	6	Total
Inputs							
Forecasted demand	6	12	18	15	13	14	78
Workforce level	6	12	18	15	13	14	78
Undertime	0	0	0	0	0	0	0
Overtime	0	0	0	0	0	0	0
Derived							
Utilized time	6	12	18	15	13	14	78
Hires	0	6	6	0	0	1	13
Layoffs	4	0	0	3	2	0	9
Calculated							
Utilized time cost	$12,000	$24,000	$36,000	$30,000	$26,000	$28,000	$156,000
Undertime cost	$0	$0	$0	$0	$0	$0	$0
Hiring cost	$0	$6,000	$6,000	$0	$0	$1,000	$13,000
Layoff cost	$2,000	$0	$0	$1,500	$1,000	$0	$4,500
Total cost	$14,000	30,000	42,000	31,500	27,000	29,000	$173,500

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b. Level Strategy

In order to minimize undertime, the maximum use of overtime possible must occur in the peak period. For this particular level strategy (other workforce options are possible), the most overtime that the manager can use is 20 percent of the regular-time capacity, w, so

$$1.20w = 18 \text{ employees required in peak period (period 3)}$$

$$w = \frac{18}{1.20} = 15 \text{ employees}$$

A 15-employee staff size minimizes the amount of undertime for this level strategy. Because the staff already includes 10 part-time employees, the manager should immediately hire five more. The complete plan is shown in Figure 15.6. The total cost is \$164,000, which seems reasonable because the minimum conceivable cost is only \$156,000 (78 periods $\times$ \$2,000/period). This cost could be achieved only if the manager found a way to cover the forecasted demand for all 78 periods with regular time. The plan seems reasonable primarily because it involves the use of large amounts of undertime (15 periods), which in this example are unpaid.

	1	2	3	4	5	6	Total
Inputs							
Forecasted demand	6	12	18	15	13	14	78
Workforce level	15	15	15	15	15	15	90
Undertime	9	3	0	0	2	1	15
Overtime	0	0	3	0	0	0	3
Derived							
Utilized time	6	12	15	15	13	14	75
Hires	5	0	0	0	0	0	5
Layoffs	0	0	0	0	0	0	0
Calculated							
Utilized time cost	\$12,000	\$24,000	\$30,000	\$30,000	\$26,000	\$28,000	\$150,000
Undertime cost	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Overtime cost	\$0	\$0	\$9,000	\$0	\$0	\$0	\$9,000
Hiring cost	\$5,000	\$0	\$0	\$0	\$0	\$0	\$5,000
Layoff cost	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total cost	\$17,000	24,000	39,000	30,000	26,000	28,000	\$164,000

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Scheduling

- Scheduling is the last step in the operations planning and scheduling process. It takes the operations and scheduling process from planning to execution, and is where the "rubber meets the road."

- It requires gathering data from sources such as demand forecasts or specific customer orders, resource availability from the sales and operations plan, and specific constraints to be reckoned with from employees and customers.

- It then involves generating a work schedule for employees or sequences of jobs or customers at workstations.

- The schedule has to be coordinated with the employees and suppliers to make sure that all constraints are satisfied.

- Here we cover Gantt charts, employee schedules, job sequencing at workstations, and software support.

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Gantt Chart

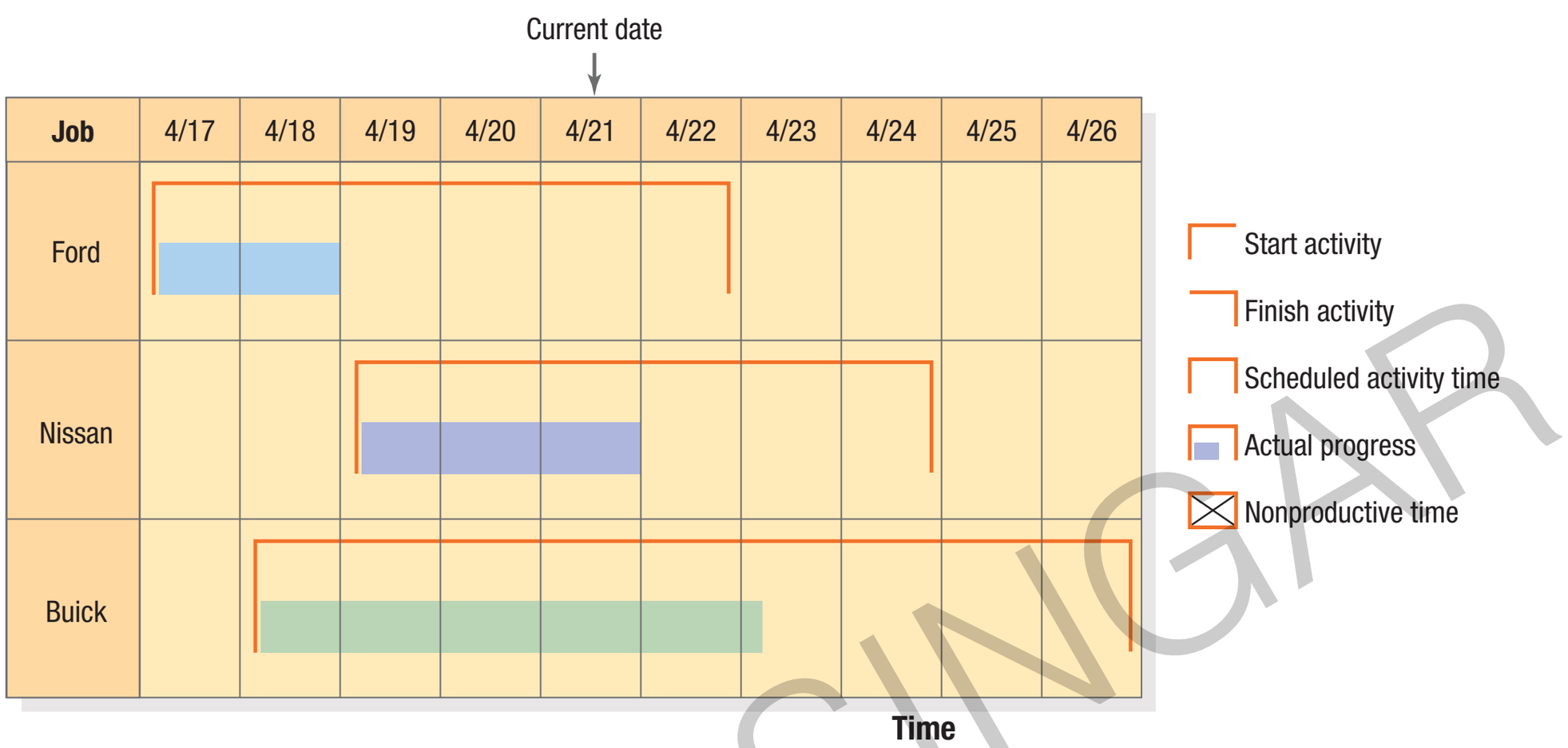
- This tool can be used to monitor the progress of work and to view the load on workstations.

- The chart takes two basic forms: (1) the job or activity progress chart and (2) the workstation chart.

1. The Gantt progress chart graphically displays the current status of each job or activity relative to its scheduled completion date. For example, suppose that an automobile parts manufacturer has three jobs under way, one each for Ford, Nissan, and Buick. The actual status of these orders is shown by the colored bars in the figure; the red lines indicate the desired schedule for the start and finish of each job.

2. The Gantt workstation chart shows the load on the work-stations and the nonproductive time. The chart can be used to identify time slots for unscheduled emergency work. It can also be used to accommodate requests to change the time of work/job. For example, Dr. Flowers may be able to change the start of her surgery to 2 PM by swapping time slots with Dr. Gillespie in operating room C or by asking Dr. Brothers to start her surgery one hour earlier in operating room A and asking Dr. Bright to schedule her surgery for the morning in operating room C.

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Workstation	7 A.M.	8 A.M.	9 A.M.	10 A.M.	11 A.M.	12 P.M.	1 P.M.	2 P.M.	3 P.M.	4 P.M.	5 P.M.	6 P.M.
Operating Room A	Dr. Jon Adams				Dr. Aubrey Brothers					Dr. Alaina Bright		
Operating Room B		Dr. Gary Case				Dr. Jeff Dow			Dr. Madeline Easton			
Operating Room C	Dr. Jordanne Flowers							Dr. Dan Gillespie				

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Employee Schedule

- Another way to manage capacity is **workforce scheduling**, which is a type of scheduling that determines when employees work. Of particular interest are situations when not all employees work the same five days a week, and same eight hours per day.

- The schedule specifies the on-duty and off-duty periods for each employee over a certain time period, as in assigning postal clerks, nurses, pilots, attendants, or police officers to specific workdays and shifts.

- This approach is used when customers demand quick response and total demand can be forecasted with reasonable accuracy. In these instances, capacity is adjusted to meet the expected loads on the service system.

- Workforce schedules translate the staffing plan into specific schedules of work for each employee.

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- The **technical constraints** imposed on the workforce schedule are the resources provided by the staffing plan and the requirements placed on the operating system. **However, other constraints, including legal and behavioral considerations, also can be imposed**. For example, Air India is required to have at least a minimum number of flight attendants on duty at all times. Similarly, a minimum number of fire and safety personnel must be on duty at a fire station at all times. Such constraints limit management's flexibility in developing workforce schedules.

- The constraints imposed by the psychological needs of workers complicate scheduling even more. Some of these constraints are written into labor agreements. For example, an employer might agree to give employees a certain number of consecutive days off per week or to limit employees' consecutive workdays to a certain maximum. Other provisions might govern the allocation of vacations, days off for holidays, or rotating shift assignments. In addition, the preferences of the employees themselves need to be considered.

- One way that managers deal with certain undesirable aspects of scheduling is to use a **rotating schedule**, which rotates employees through a series of workdays or hours. Thus, over a period of time, each person has the same opportunity to have weekends and holidays off and to work days, as well as evenings and nights. A rotating schedule gives each employee the next employee's schedule the following week. In contrast, a **fixed schedule** calls for each employee to work the same days and hours each week.

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Example: Developing a Workforce Schedule

The Amalgamated Parcel Service is open seven days a week. The schedule of requirements is

Day	M	T	W	Th	F	S	Su
Required number of employees	6	4	8	9	10	3	2

The manager needs a workforce schedule that provides two consecutive days off and minimizes the amount of total slack capacity. To break ties in the selection of off days, the scheduler gives preference to Saturday and Sunday if it is one of the tied pairs. If not, she selects one of the tied pairs arbitrarily.

SOLUTION

Friday contains the maximum requirements, and the pair S–Su has the lowest total requirements. Therefore, Employee 1 is scheduled to work Monday through Friday.

Note that Friday still has the maximum requirements and that the requirements for the S–Su pair are carried forward because these are Employee 1's days off. These updated requirements are the ones the scheduler uses for the next employee.

The day-off assignments for the employees are shown in the following table.

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M	T	W	Th	F	S	Su	Employee	Comments
6	4	8	9	10	3	2	**1**	The S–Su pair has the lowest total requirements. Assign Employee **1** to a Monday through Friday schedule and update the requirements.
5	3	7	8	9	3	2	**2**	The S–Su pair has the lowest total requirements. Assign Employee **2** to a Monday through Friday schedule and update the requirements.
4	2	6	7	8	3	2	**3**	The S–Su pair has the lowest total requirements. Assign Employee **3** to a Monday through Friday schedule and update the requirements.
3	1	5	6	7	3	2	**4**	The M–T pair has the lowest total requirements. Assign Employee **4** to a Wednesday through Sunday schedule and update the requirements.
3	1	4	5	6	2	1	**5**	The S–Su pair has the lowest total requirements. Assign Employee **5** to a Monday through Friday schedule and update the requirements.
2	0	3	4	5	2	1	**6**	The M–T pair has the lowest total requirements. Assign Employee **6** to a Wednesday through Sunday schedule and update the requirements.
2	0	2	3	4	1	0	**7**	The S–Su pair has the lowest total requirements. Assign Employee **7** to a Monday through Friday schedule and update the requirements.
1	0	1	2	3	1	0	**8**	Four pairs have the minimum requirement and the lowest total: S–Su, Su–M, M–T, and T–W. Choose the S–Su pair according to the tie-breaking rule. Assign Employee **8** to a Monday through Friday schedule and update the requirements.
0	0	0	1	2	1	0	**9**	Arbitrarily choose the Su–M pair to break ties because the S–Su pair does not have the lowest total requirements. Assign Employee **9** to a Tuesday through Saturday schedule and update the requirements.
0	0	0	0	1	0	0	**10**	Choose the S–Su pair according to the tie-breaking rule. Assign Employee **10** to a Monday through Friday schedule.

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FINAL SCHEDULE

Employee	M	T	W	Th	F	S	Su	Total
1	X	X	X	X	X	off	off	
2	X	X	X	X	X	off	off	
3	X	X	X	X	X	off	off	
4	off	off	X	X	X	X	X	
5	X	X	X	X	X	off	off	
6	off	off	X	X	X	X	X	
7	X	X	X	X	X	off	off	
8	X	X	X	X	X	off	off	
9	off	X	X	X	X	X	off	
10	X	X	X	X	X	off	off	
Capacity, C	7	8	10	10	10	3	2	50
Requirements, R	6	4	8	9	10	3	2	42
Slack, C – R	1	4	2	1	0	0	0	8

DECISION POINT

With its substantial amount of slack capacity, the schedule is not unique. Employee 9, for example, could have Sunday and Monday, Monday and Tuesday, or Tuesday and Wednesday off without causing a capacity shortage. Indeed, the company might be able to get by with one fewer employee because of the total of eight slack days of capacity. However, all 10 employees are needed on Fridays. If the manager were willing to get by with only nine employees on Fridays or if someone could work one day of overtime on a rotating basis, he would not need Employee 10. As indicated in the table, the net requirement left for Employee 10 to satisfy amounts to only one day, Friday. Thus, Employee 10 can be used to fill in for vacationing or sick employees.

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Job Sequencing at Work Stations

- Another aspect of scheduling is sequencing work at workstations. **Sequencing** determines the order in which jobs or customers are processed in the waiting line at a workstation. When combined with the expected processing time, the sequence allows you to estimate the start and finish times of each job.

- One way to determine what job or customer to process next is with the help of a **priority sequencing rule**. The following two priority sequencing rules are commonly used in practice.

1. **First-Come, First-Served:** The job or customer arriving at the workstation first has the highest priority under a first-come, first-served (FCFS) rule. This rule is the most "democratic" in that each job is treated equally, with no one stepping ahead of others already in line.

2. **Earliest Due Date:** The job or customer with the earliest due date (EDD) is the next one to be processed. The due date specifies when work on a job or customer should be finished. Due dates are commonly used by manufacturers and suppliers in the supply chain. For example, a product cannot be assembled until all of its purchased and produced components are available. If these components were not already in inventory, they must be ordered prior to when the product assembly can begin. Their due date is the start date for assembling the product to be assembled. It is also the key to **expediting**, which is the process of completing a job sooner than would otherwise be done.

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- **Performance Measures:** The quality of a schedule can be judged in various ways. Two commonly used performance measures are flow time and past due.

1. **Flow Time:** The amount of time a job spends in the service or manufacturing system is called flow time. It is the sum of the waiting time for servers or machines; the process time, including setups; the time spent moving between operations; and delays resulting from machine breakdowns, unavailability of facilitating goods or components, and the like. Flow time is sometimes referred to as *throughput time* or *time spent in the system, including service*. For a set of jobs to be processed at a single workstation. a job's flow time is Flow time = Finish time + Time since job arrived at workstation.

2. **Past Due:** The measure past due can be expressed as the amount of time by which a job missed its due date (also referred to as tardiness) or as the percentage of total jobs processed over some period of time that missed their due dates. Minimizing these past due measures supports the competitive priorities of cost (penalties for missing due dates), quality (perceptions of poor service), and time (on-time delivery).

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Example: Using the Two Priority Sequencing Rules

Currently a consulting company has five jobs in its backlog. The time since the order was placed, processing time, and promised due dates are given in the following table. Determine the schedule by using the FCFS rule, and calculate the average days past due and flow time. How can the schedule be improved, if average flow time is the most critical?

Customer	Time Since Order Arrived (days ago)	Processing Time (days)	Due Date (days from now)
A	15	25	29
B	12	16	27
C	5	14	68
D	10	10	48
E	0	12	80

SOLUTION

a. The FCFS rule states that Customer A should be the first one in the sequence, because that order arrived earliest—15 days ago. Customer E's order arrived today, so it is processed last. The sequence is shown in the following table, along with the days past due and flow times.

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Customer Sequence	Start Time (days)		Processing Time (days)		Finish Time (days)	Due Date	Days Past Due	Days Ago Since Order Arrived	Flow Time (days)
A	0	+	25	=	25	29	**0**	15	**40**
B	25	+	16	=	41	27	**14**	12	**53**
D	41	+	10	=	51	48	**3**	10	**61**
C	51	+	14	=	65	68	**0**	5	**70**
E	65	+	12	=	77	80	**0**	0	**77**

The *finish time* for a job is its start time plus the processing time. Its finish time becomes the start time for the next job in the sequence, assuming that the next job is available for immediate processing. The days past due for a job is zero (0) if its due date is equal to or exceeds the finish time. Otherwise it equals the shortfall. The flow time for each job equals its finish time plus the number of days ago since the order first arrived at the workstation. For example, Customer C's flow time is its scheduled finish time of 65 days plus the 5 days since the order arrived, or 70 days. The days past due and average flow time performance measures for the FCFS schedule are

$$\text{Average days past due} = \frac{0 + 14 + 3 + 0 + 0}{5} = 3.4 \text{ days}$$

$$\text{Average flow time} = \frac{40 + 53 + 61 + 70 + 77}{5} = 60.2 \text{ days}$$

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b. The average flow time can be reduced. One possibility is the sequence shown in the following table, which uses the Shortest Processing Time (SPT) rule, which is one of several rules developed more fully in MyOMLab Supplement J, "Operations Scheduling." (For still another possibility, see Solved Problem 1, which applies the EDD rule.)

MyOMLab

Customer Sequence	Start Time (days)		Processing Time (days)		Finish Time (days)	Due Date	Days Past Due	Days Ago Since Order Arrived	Flow Time (days)
D	0	+	10	=	10	48	**0**	10	**20**
E	10	+	12	=	22	80	**0**	0	**22**
C	22	+	14	=	36	68	**0**	5	**41**
B	36	+	16	=	52	27	**25**	12	**64**
A	52	+	25	=	77	29	**48**	15	**92**

$$\text{Average days past due} = \frac{0 + 0 + 0 + 25 + 48}{5} = 14.6 \text{ days}$$

$$\text{Average flow time} = \frac{20 + 22 + 41 + 64 + 92}{5} = 47.8 \text{ days}$$

This schedule reduces the average flow time from 60.2 to 47.8 days—a 21 percent improvement. However, the past due times for jobs A and B have increased.

DECISION POINT

Management decided to use a modified version of the second schedule, adding overtime when Customer B is processed. Further, Customer A agreed to extend its due date to 77 days, because in this case the advanced warning allowed it to reschedule its own operations with little problem.

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Software Support

- Computerized scheduling systems are available to cope with the complexity of workforce scheduling. In some types of firms, such as telephone companies, mail-order catalog houses, or emergency hotline agencies, employees must be on duty 24 hours a day, 7 days a week.

- Management must consider the timing of lunch breaks and rest periods, the number and starting times of shift schedules, and the days off for each employee. The programs select the schedule that minimizes the sum of expected costs of over and understaffing.

- Software is also available for sequencing jobs at workstations. They help firms design and manage the linkages between customers and suppliers in the supply chain. Coupled with the Internet and improved data storage and manipulation methods, such computer software has given rise to **advanced planning and scheduling (APS) systems**, which seek to optimize resources across the supply chain and align daily operations with strategic goals. A firm's ability to change its schedules quickly and still keep the goods and services flowing smoothly through the supply chain provides a competitive edge.

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